

Montserrat Arriaga Santana

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Education

Universidad Nacional Autónoma de México (UNAM)	<i>Mexico City, Mexico</i>
B.S. in Computer Science	Sept. 2021 – Present
Universidad Nacional Autónoma de México (UNAM)	<i>Mexico City, Mexico</i>
B.A. in English Language	Sept. 2023 – Present

Experience

Honeywell Technology Solutions	Jan. 2024 – Apr. 2025
<i>Software Cybersecurity Engineer</i>	<i>SVN, C++, Kotlin, Azure, Black Duck</i>
- Automated internal FieldSense-related workflows at the research center, streamlining processes and improving operational efficiency. - Implemented cybersecurity measures for UDP-based communication, supporting the protection of sensitive data and applications. - Contributed to strengthening the organization's cybersecurity framework through software-security analysis and engineering support.	
AWL Co.	
<i>Software Engineer Intern</i>	<i>Python, MobaXTerm, Tegrastats</i>
- Selected as one of the few Mexican participants in a global computational development and artificial intelligence program organized by Japan's Ministry of Economy. - Developed the initial version of a performance monitoring tool for AWLEngineAgentService using Python. - Automated the collection of CPU, GPU, memory, and FPS metrics, generating customizable reports in JSON, CSV, and PDF formats.	
National University of Singapore	Jun. 2025 – Aug. 2025
<i>Research Intern</i>	<i>Python, JavaScript, GitHub, NLP</i>
- Selected for a competitive NUS research internship program in computer science. - Conducted large-scale text analysis on political content using NLP techniques, transformer-based embeddings, and clustering algorithms. - Built and optimized pipelines for tweet cleaning, embedding generation, dimensionality reduction, and unsupervised clustering. - Applied cosine-normalized embeddings and HDBSCAN-based clustering to identify topic patterns and discourse shifts.	
Linguistic Engineering Group (UNAM)	Ongoing
<i>Research Assistant</i>	<i>Python, NLP, Vision-Language Models, QLoRA</i>
- Contributed to unpaid research projects in NLP and multimodal learning at UNAM's Linguistic Engineering Group. - Designed the QLoRA fine-tuning strategy for Qwen2.5-VL-72B in a published CLEF 2026 Working Notes paper on sexism detection in memes. - Built task-specific prompt formats and JSON-based output schemas for sexism detection, source-intention classification, and multi-label sexism categorization. - Supported the multimodal classification pipeline, including model adaptation, inference, validation, and official shared-task evaluation analysis.	

Selected Publications & Research Projects

Detecting Sexism in Memes with Vision-Language Models for EXIST 2026

CLEF 2026 Working Notes

Co-author, UNAM_GIL Lab

Vision-Language Models, QLoRA, NLP, Multimodal Classification

- Designed the fine-tuning approach for Qwen2.5-VL-72B using QLoRA for multimodal sexism detection in memes.
- Contributed to the experimental design for binary sexism detection, source-intention classification, and multi-label sexism categorization.
- Helped evaluate model performance across Spanish, English, and multilingual settings using official shared-task metrics.

Towards Benchmarking Western Bluebird Detection in the Wild Accepted: LatinX in CVPR 2026; CVBMC in ICPR 2026

Co-first Author

Computer Vision, Object Detection, Segmentation, Benchmarking

- Designed and curated a benchmark for Western Bluebird detection and segmentation in real-world ecological video frames.
- Manually labeled and verified bird instances, including bounding boxes and segmentation masks, for small-object detection under natural conditions.
- Evaluated supervised, transformer-based, open-vocabulary, and segmentation models, including YOLO, Faster R-CNN, RT-DETR, YOLO-World, Grounding DINO, and SAM-based methods.

Water Echoes

Jul. 2023

NASA Space Apps Challenge Project

Spring Boot, PostgreSQL, JavaScript, React Native, Git

- Built an application that uses NASA open data to geo-localize nearby water bodies and inform users about endangered species.
- Placed among the top 6 projects out of 22 teams in the NASA Space Apps Challenge hackathon.
- Designed and implemented the Spring Boot API architecture and relational database model.

Skills

Programming Languages:

Python, Java, C, C++, SQL, HTML, JavaScript, Typescript

Languages:

English (C2), Spanish

Technologies & Tools:

Git, Linux, Docker

Certifications:

Java Programación Orientada a Objetos, Software Security Practitioner,

The Psychology of Games - Secrets of Good Game Design, The Ultimate Guide to Game Development with Unity .